

Student Number	
Mark / 64	

Chemistry

**Final Examination
Preliminary Course • 2003**

General Instructions

- Reading time – 5 minutes
- Working time – 120 minutes
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- A Data Sheet and a Periodic Table are provided at the back of this paper
- Write your Student Number at the top of this page

Total Marks – 64

Part A – 10 marks

- Attempt Questions 1 – 10
- Allow about 10 minutes for this part

Part B – 54 marks

- Attempt Questions 11 – 23
- Allow about 110 minutes for this part

Part A – 10 marks

Attempt Questions 1–10

Allow about 10 minutes for this part

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct ↗

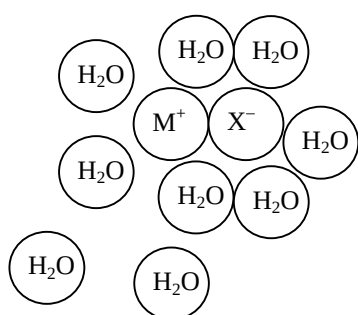
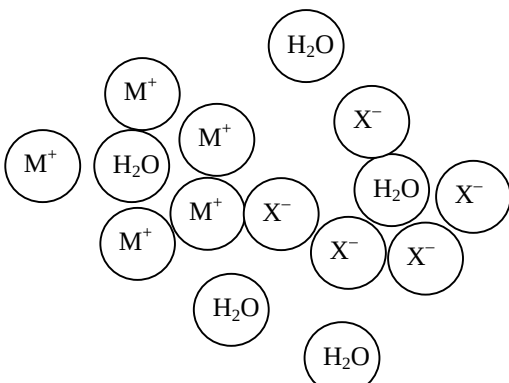
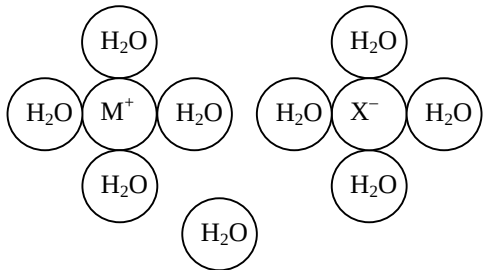
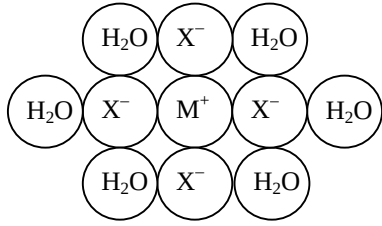
Answer Box for Questions 1–10				
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3	A ☐	B ☐	C ☐	D ☐
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► Mark your answers for Questions 1 – 10 in the Answer Box on page 1.

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- (A) ammonia
- (B) carbon monoxide
- (C) soot
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2 Which diagram shows the complete dissolution of an ionic solid (M^+X^-) in water?

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	sphere	percentage water
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- (A) The final energy content of the system is greater than the initial energy content.
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5 Organisms living in an aquatic habitat experience less temperature extremes than nearby organisms living on the land. Which factor explains the moderating effect of the water?

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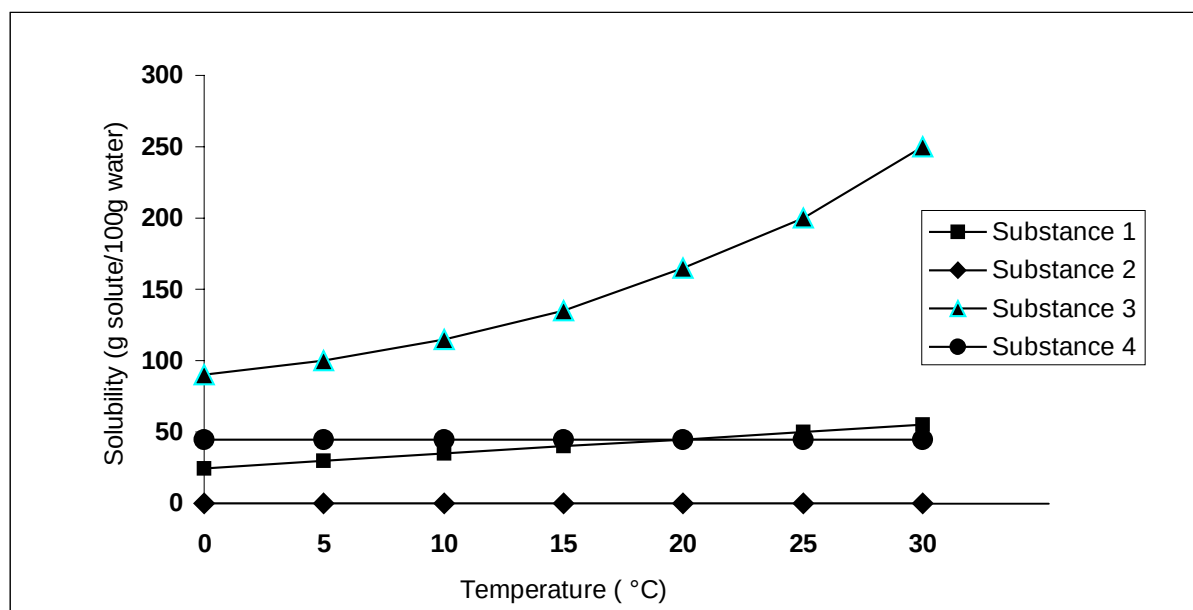
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Which substance would be a covalent network solid?

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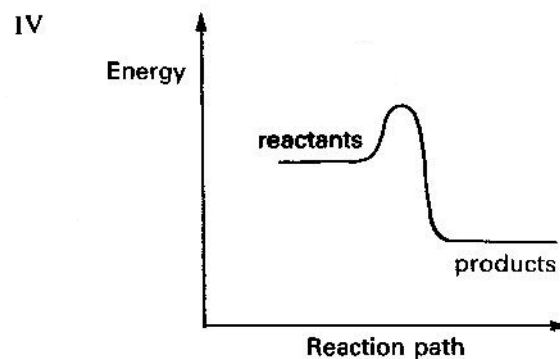
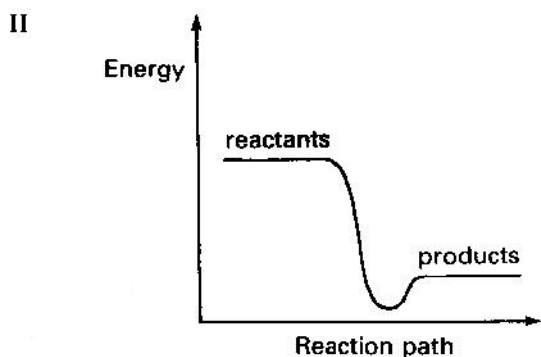
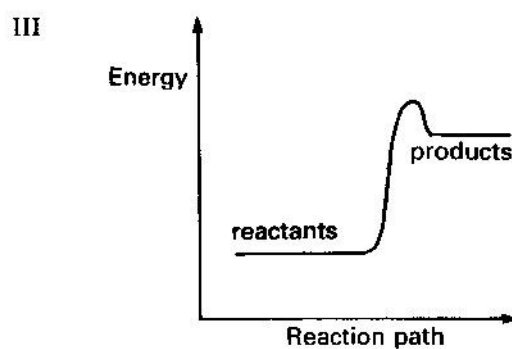
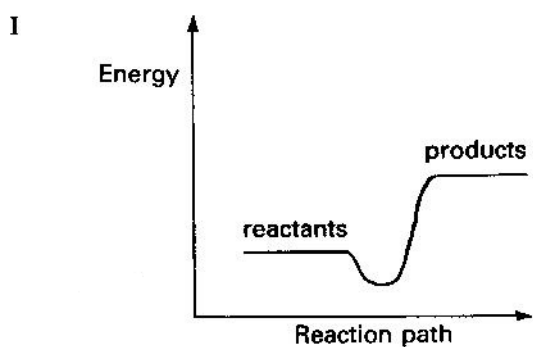
Attempt Questions 11 – 23

Allow about 110 minutes for this part

► **Show all relevant working in questions involving calculations.**

Question 11 (3 marks)

The graphs show the energy changes during the course of four different situations (I – IV)...



- (a) Which graph could correspond to the reaction: $\text{NO}_2(\text{g}) \rightarrow \frac{1}{2}\text{N}_2(\text{g}) + \text{O}_2(\text{g})$ $\Delta H = -33.7 \text{ kJ mol}^{-1}$?

_____ (1 mark)

- (b) Which graph could correspond to the melting of ice? _____ (1 mark)

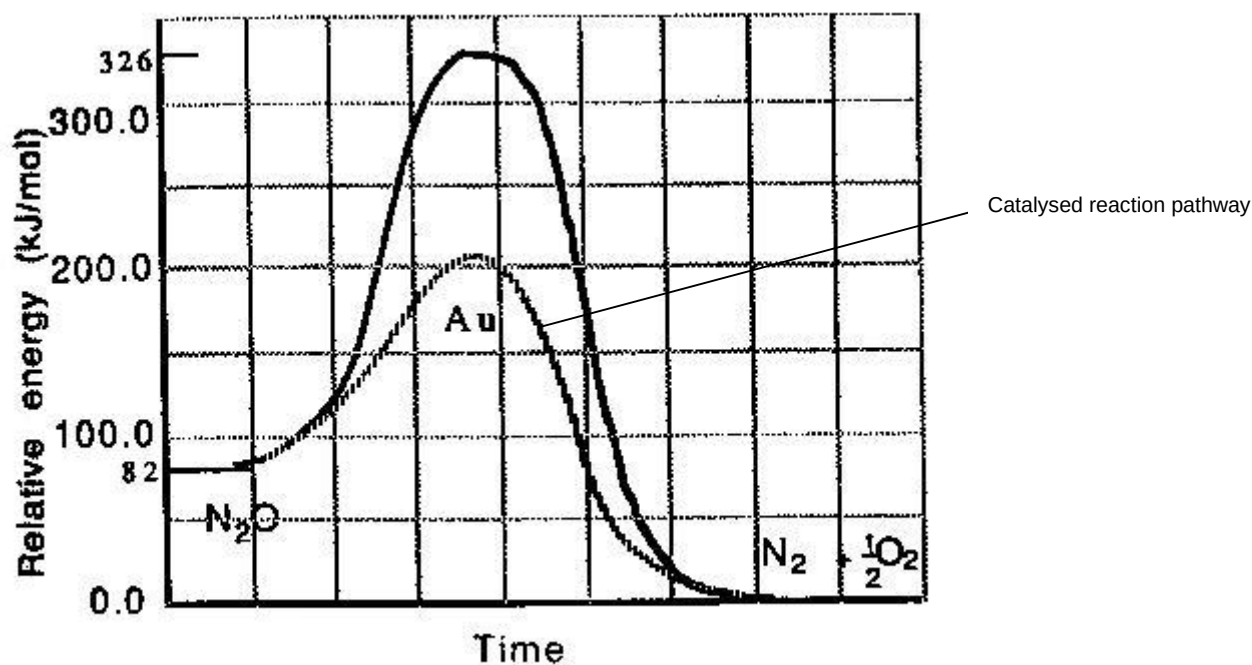
- (c) Which graph could correspond to the combustion of methane? _____ (1 mark)

Question 12 (4 marks)

- (a) Explain how fine coal dust in a coal mine can be an explosive hazard. **(2 marks)**

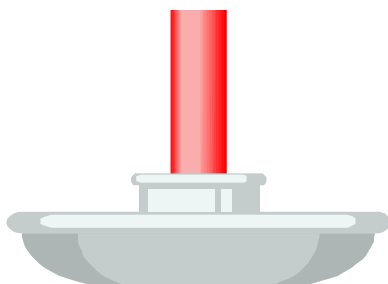
- (b) Suggest one safety feature adopted by industries to avoid dust explosions. **(1 mark)**

- (c) Dinitrogen monoxide can be thermally decomposed to nitrogen and oxygen.
The reaction is catalysed by gold... $\text{N}_2\text{O (g)} \rightarrow \text{N}_2 \text{(g)} + \frac{1}{2}\text{O}_2 \text{(g)} \quad \Delta H = -82 \text{ kJ mol}^{-1}$



What is the activation energy for the reverse reaction... $\text{N}_2 \text{(g)} + \frac{1}{2}\text{O}_2 \text{(g)} \rightarrow \text{N}_2\text{O (g)}$
in the absence of the gold catalyst? **(1 mark)**

Question 13 (3 marks)



A candle without a wick will not burn



A candle with a wick will burn readily

- (a) Identify two physical changes occurring during the burning of a candle. **(2 marks)**

- (b) Explain why only the candle with the wick burns. **(1 mark)**

Question 14 (6 marks)

- (a) Construct the Lewis electron dot structures for ammonia and water. **(2 marks)**

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Question 14 continues on page 8.

Question 14 (continued)

- (b) Describe the shape of the hydrogen sulfide molecule and explain why hydrogen sulfide has this shape. **(2 marks)**

- (c) Draw a diagram of an ammonia molecule showing its correct shape. **(1 mark)**
Identify the shape. **(1 mark)**



Question 15 (3 marks)

- (a) Which of hydrogen sulfide and water has the higher boiling point? **(1 mark)**

- (b) Explain your answer to (a). **(2 marks)**

Question 16 (3 marks)

- (a) What property of water enables an insect to walk on water? **(1 mark)**

- (b) Explain the nature of the property of water identified in (a) in terms of intermolecular forces. **(2 marks)**

Question 17 (3 marks)

You have already done an experiment where you used water's ability to absorb heat to measure energy changes in reactions.

- (a) Identify or describe the apparatus you used to perform the experiment. **(1 mark)**

- (b) Identify one measurement required to do this experiment. **(1 mark)**

- (c) Write the equation you used to determine the amount of heat absorbed. **(1 mark)**

Question 18 (5 marks)

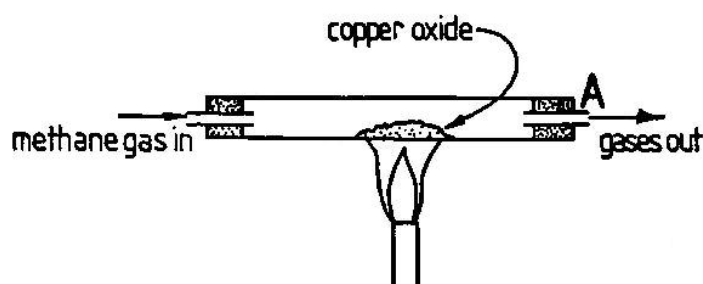
A cargo helicopter accidentally dropped 1200 kg of chemical in a pond containing 50,000 litres of water. When the chemical dissolved in the pond, the temperature increased.

- (a) How much heat was released to the water in the pond, if the water temperature in the pond increased from 15°C to 21°C? **(2 marks)**

- (b) Outline three implications (other than directly killing organisms) for aquatic life subjected to thermal pollution. **(3 marks)**

Question 19 (5 marks)

The diagram shows methane gas passing over heated copper(II) oxide reacting to produce copper metal and gaseous products of carbon dioxide and water vapour which leave the apparatus at A...



- (a) Write the balanced formulae equation for the reaction of copper(II) oxide with methane (CH_4). **(1 mark)**

- (b) If 15.9 g of copper(II) oxide is completely reacted, calculate the mass of copper metal formed. **(2 marks)**

- (c) Calculate the percentage of copper in the copper(II) oxide sample. **(1 mark)**

- (d) Calculate the volume of CO_2 produced at 25°C and 100 kPa from 15.9 g of copper(II) oxide. **(1 mark)**

Question 20 (4 marks)

- (a) Legislation states that the concentration of alcohol, $\text{C}_2\text{H}_5\text{OH}$, in the blood of an experienced driver of a motor car is not to exceed 0.05% (w/v).

Calculate the corresponding $\text{C}_2\text{H}_5\text{OH}$ concentration in the blood in moles per litre. **(2 marks)**

- (b) Identify a different measurement of concentration to that mentioned above (i.e. w/v) and describe a use for this measurement. **(2 marks)**

Question 21 (4 marks)

- (a) Calculate the mass of sodium sulfate required to prepare 50.0 mL of 0.150 mol L^{-1} solution. **(1 mark)**

- (b) What volume of this solution would you need to dilute, to prepare 125 mL of $0.0500 \text{ mol L}^{-1}$ solution? **(1 mark)**

- (c) What is the concentration of the sodium ions and sulfate ions in the $0.0500 \text{ mol L}^{-1}$ solution? **(2 marks)**

Question 22 (4 marks)

The table shows data for the compound hydrazine...

Composition	Hydrazine is a compound of nitrogen and hydrogen
Complete combustion of gaseous hydrazine at 400 K and 100 kPa	$\begin{array}{ccccccc} \text{hydrazine (g)} & + & \text{oxygen (g)} & \rightarrow & \text{nitrogen dioxide (g)} & + & \text{water (g)} \\ 1.0 \text{ L} & & 3.0 \text{ L} & & 2.0 \text{ L} & & 2.0 \text{ L} \end{array}$

- (a) Explain how the data for combustion illustrates Gay-Lussac's Law of Combining Gas Volumes. **(1 mark)**

- (b) Determine the molecular formula of hydrazine.
Show all working. **(2 marks)**

- (c) What is the empirical formula of hydrazine? **(1 mark)**

Question 23 (7 marks)

2.08 g of barium chloride was dissolved in water to make 50.0 mL of solution and then added to 50.0 mL of an aqueous solution containing 2.84 g of sodium sulfate. A white precipitate formed.

- (a) Write the net ionic equation for the reaction forming the precipitate. **(1 mark)**
► Use the solubility table below to determine the identity of the precipitate.

ANION	CATION	COMPOUND
All	Group I metals	soluble
All	Ammonium, NH_4^+	soluble
Nitrate, NO_3^-	All	soluble
Acetate/ethanoate CH_3COO^-	All except Ag^+	soluble
Chloride, Cl^- Bromide, Br^- Iodide, I^-	Ag^+ , Pb^{2+} , Hg_2^{2+} , Cu^+	insoluble
	All others	soluble
Sulfate, SO_4^{2-}	Ca^{2+} , Sr^{2+} , Ba^{2+} , Pb^{2+} , Ag^+ , Hg_2^{2+}	insoluble
	All others	soluble
Sulfide, S^{2-}	Group I and II metals, NH_4^+	soluble
	All others	insoluble
Hydroxide, OH^-	Group I metals, NH_4^+ , Sr^{2+} , Ba^{2+}	soluble
	All others	insoluble
Carbonate, CO_3^{2-} Phosphate, PO_4^{3-} Sulfite, SO_3^{2-}	Group I metals, NH_4^+	soluble
	All others	insoluble

Question 23 continues on page 15

Question 23 (continued)

(b) What is the mass of the precipitate formed? Show working. **(4 marks)**

(c) Calculate the concentration (mol L^{-1}) of sulfate ions in the final solution. Show working. **(2 marks)**

HIGHER SCHOOL CERTIFICATE EXAMINATION
Chemistry

DATA SHEET

Avogadro constant, N_A	$6.022 \times 10^{23} \text{ mol}^{-1}$
Volume of 1 mole ideal gas: at 100 kPa and	
at 0°C (273.15 K)	22.71 L
at 25°C (298.15 K)	24.79 L
Ionisation constant for water at 25°C (298.15 K), K_w	1.0×10^{-14}
Specific heat capacity of water	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$

Some useful formulae

$$\text{pH} = -\log_{10}[\text{H}^+] \qquad \Delta H = -m C \Delta T$$

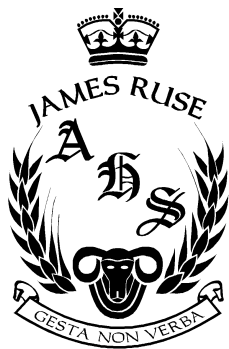
Some standard potentials

$\text{K}^+ + \text{e}^-$	$\rightleftharpoons$	K(s)	-2.94 V
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ba(s)	-2.91 V
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ca(s)	-2.87 V
$\text{Na}^+ + \text{e}^-$	$\rightleftharpoons$	Na(s)	-2.71 V
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mg(s)	-2.36 V
$\text{Al}^{3+} + 3\text{e}^-$	$\rightleftharpoons$	Al(s)	-1.68 V
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Mn(s)	-1.18 V
$\text{H}_2\text{O} + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g}) + \text{OH}^-$	-0.83 V
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Zn(s)	-0.76 V
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Fe(s)	-0.44 V
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Ni(s)	-0.24 V
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Sn(s)	-0.14 V
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Pb(s)	-0.13 V
$\text{H}^+ + \text{e}^-$	$\rightleftharpoons$	$\frac{1}{2}\text{H}_2(\text{g})$	0.00 V
$\text{SO}_4^{2-} + 4\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	$\text{SO}_2(\text{aq}) + 2\text{H}_2\text{O}$	0.16 V
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightleftharpoons$	Cu(s)	0.34 V
$\frac{1}{2}\text{O}_2(\text{g}) + \text{H}_2\text{O} + 2\text{e}^-$	$\rightleftharpoons$	2OH^-	0.40 V
$\text{Cu}^+ + \text{e}^-$	$\rightleftharpoons$	Cu(s)	0.52 V
$\frac{1}{2}\text{I}_2(\text{s}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.54 V
$\frac{1}{2}\text{I}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	I^-	0.62 V
$\text{Fe}^{3+} + \text{e}^-$	$\rightleftharpoons$	Fe^{2+}	0.77 V
$\text{Ag}^+ + \text{e}^-$	$\rightleftharpoons$	Ag(s)	0.80 V
$\frac{1}{2}\text{Br}_2(\text{l}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.08 V
$\frac{1}{2}\text{Br}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Br^-	1.10 V
$\frac{1}{2}\text{O}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^-$	$\rightleftharpoons$	H_2O	1.23 V
$\frac{1}{2}\text{Cl}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.36 V
$\frac{1}{2}\text{Cr}_2\text{O}_7^{2-} + 7\text{H}^+ + 3\text{e}^-$	$\rightleftharpoons$	$\text{Cr}^{3+} + \frac{7}{2}\text{H}_2\text{O}$	1.36 V
$\frac{1}{2}\text{Cl}_2(\text{aq}) + \text{e}^-$	$\rightleftharpoons$	Cl^-	1.40 V
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^-$	$\rightleftharpoons$	$\text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.51 V
$\frac{1}{2}\text{F}_2(\text{g}) + \text{e}^-$	$\rightleftharpoons$	F^-	2.89 V

Aylward and Findlay, *SI Chemical Data* (5th Edition) is the principal source of data for this examination paper. Some data may have been modified for examination purposes.

PERIODIC TABLE OF THE ELEMENTS

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1																He	
H 1.008 Hydrogen																4.003 Helium	
KEY																	
		Atomic Number		Atomic Weight		Symbol of element											
		79		197.0		Au		Gold		Name of element							



ANSWERS and MARKING SCHEME

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A  B  C  D 

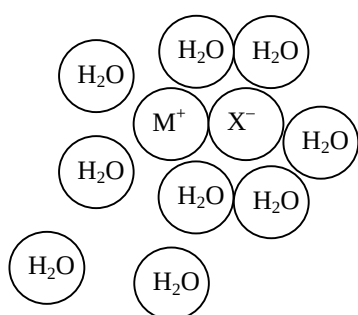
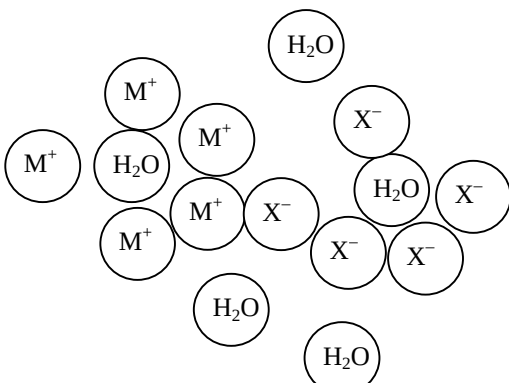
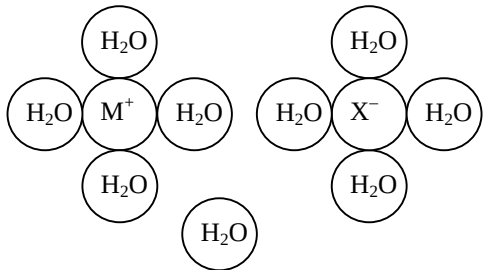
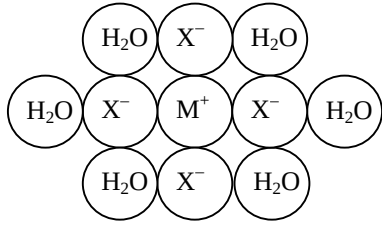
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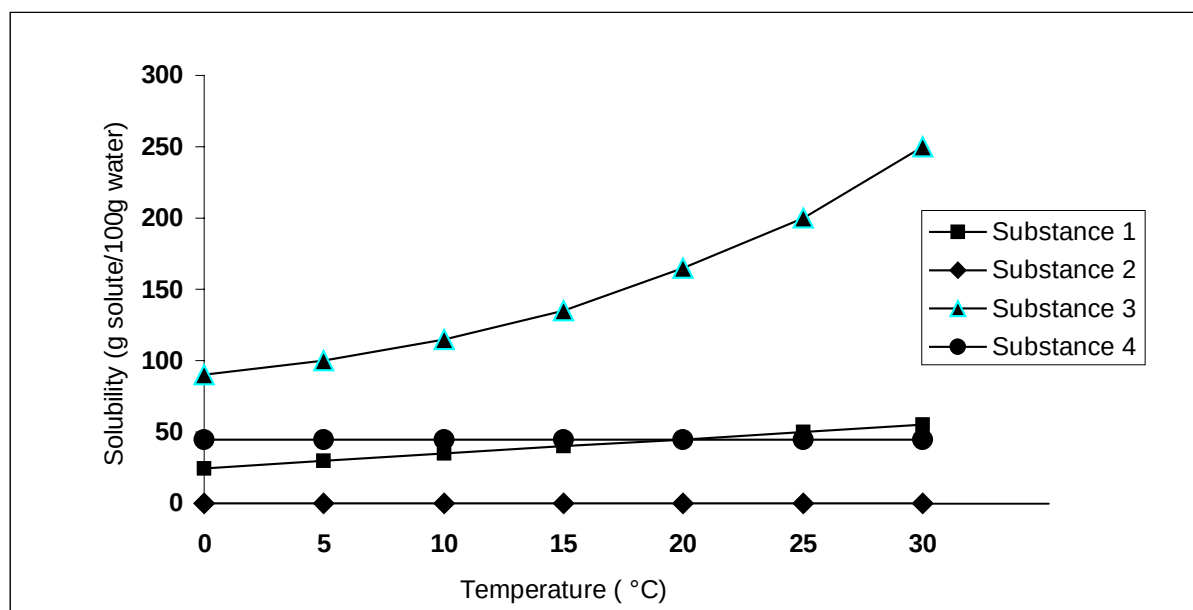
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Part B – 54 marks

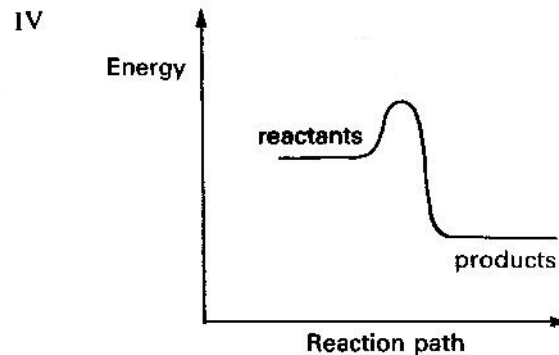
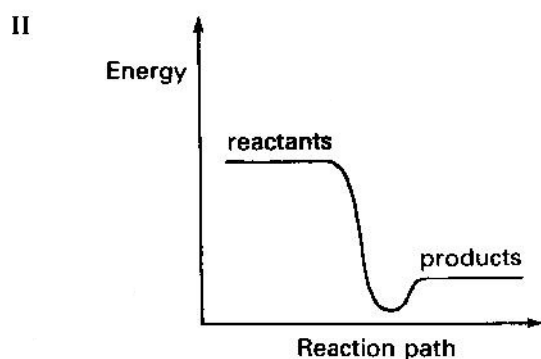
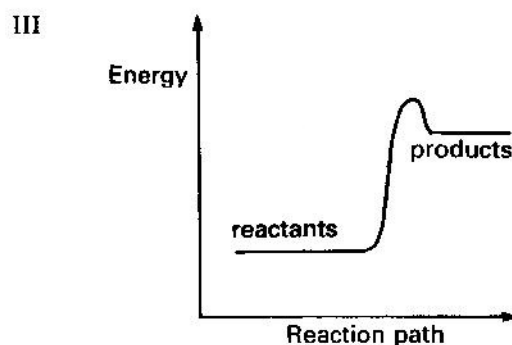
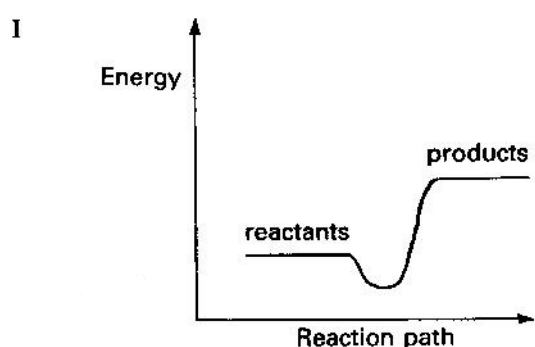
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Question 11 O/C – P7 (3 marks)

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Graph IV (1 mark)

- (b) Which graph could correspond to the melting of ice? Graph III (1 mark)

- (c) Which graph could correspond to the combustion of methane? Graph IV (1 mark)

Question 12 (4 marks)

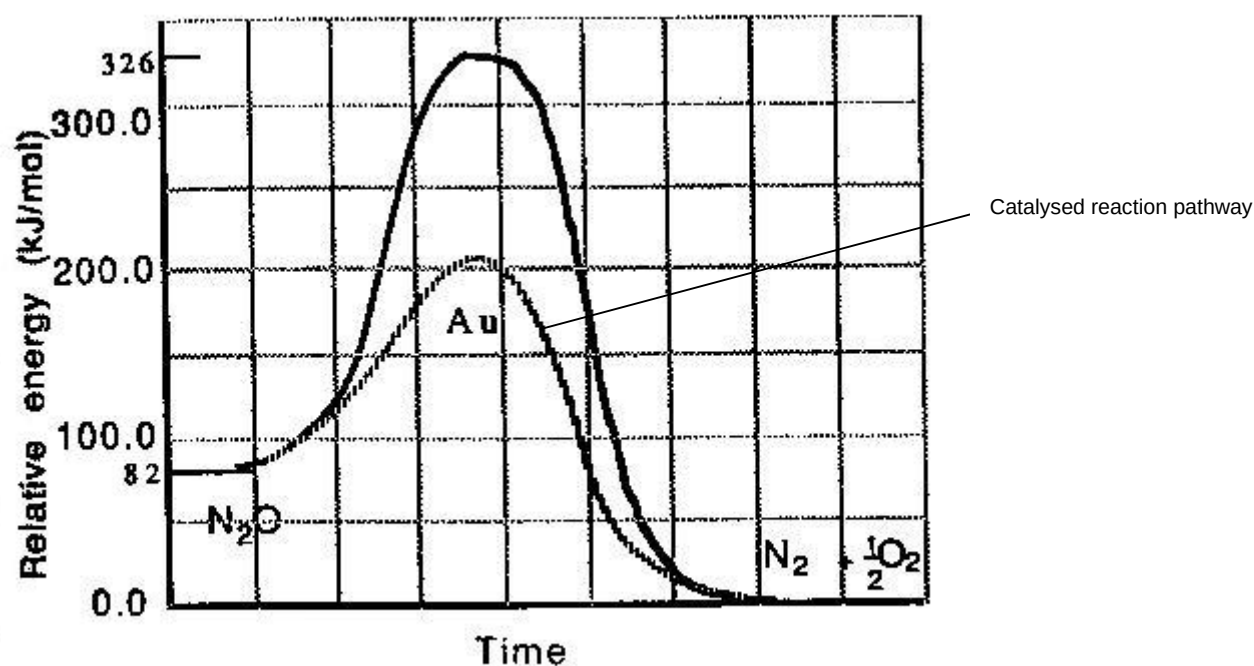
- (a) Explain how fine coal dust in a coal mine can be an explosive hazard. **O/C – P4 (2 marks)**

The total surface area of the particles is large (1 mark) and each particle has a ready supply of oxygen available. As the reaction is exothermic (1 mark), heat released supplies the activation energy to hasten further reactions and an explosion ensues.

- (b) Suggest one safety feature adopted by industries to avoid dust explosions. **O/C – P4 (1 mark)**

Ensuring there is no build-up of fine flammable particles by well-ventilating the worksite.

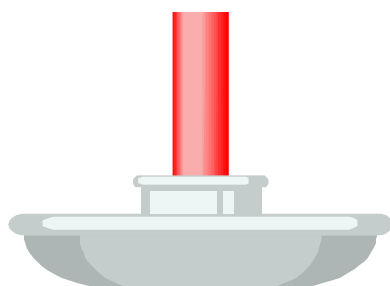
- (c) Dinitrogen monoxide can be thermally decomposed to nitrogen and oxygen.
The reaction is catalysed by gold... $\text{N}_2\text{O}_{(g)} \rightarrow \text{N}_{2(g)} + \frac{1}{2}\text{O}_{2(g)} \quad \Delta H = -82 \text{ kJ mol}^{-1}$



What is the activation energy for the reverse reaction... $\text{N}_{2(g)} + \frac{1}{2}\text{O}_{2(g)} \rightarrow \text{N}_2\text{O}_{(g)}$
in the absence of the gold catalyst? **O/C – P7, 14 (1 mark)**

326 kJ

Question 13 O/C – P7 (3 marks)



A candle without a wick will not burn



A candle with a wick will burn readily

- (a) Identify two physical changes occurring during the burning of a candle. **(2 marks)**

Melting or fusion or solid → liquid

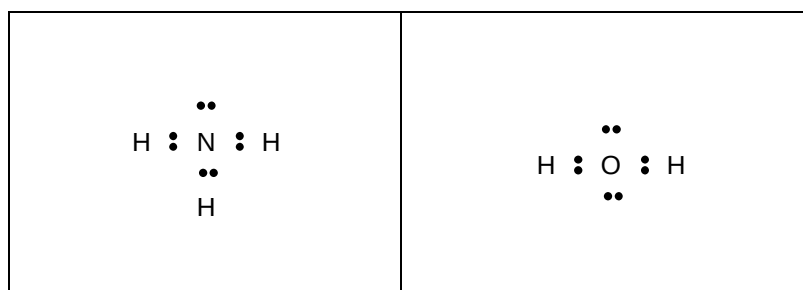
Vaporisation or liquid → gas

- (b) Explain why only the candle with the wick burns. **(1 mark)**

Wax cannot burn directly as a solid. A burning wick provides heat to melt and vapourise the wax which can then be fed up the wick (by capillary action) to sustain the flame and hence continue the combustion process.

Question 14 (6 marks)

- (a) Construct the Lewis electron dot structures for ammonia and water. **O/C – P7 (2 marks)**



Question 14 (continued)

- (b) Describe the shape of the hydrogen sulfide molecule and explain why hydrogen sulfide has this shape. **O/C – P14 (2 marks)**

Hydrogen sulfide is a bent molecule (1 mark) due to the central sulfur atom having two lone (non-bonding) pairs of electrons. This causes repulsion of the covalent bonds (1 mark).

► *The four pairs of electrons around the central sulfur atom will automatically orientate towards maximum repulsion resulting in a tetrahedral placement in 3D space. However, the shape of a molecule is determined by the placement of the bonded atoms, ∴ in this case, bent.*

- (c) Draw a diagram of an ammonia molecule showing its correct shape. **(1 mark) O/C – P14**
Identify the shape. **(1 mark)**

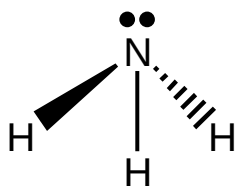


Diagram (1 mark)

Ammonia is a trigonal pyramid (1 mark)

► 'pyramidal' accepted as per Smith text

Question 15 (3 marks)

- (a) Which of hydrogen sulfide and water has the higher boiling point? **O/C – P14 (1 mark)**

Water has the higher boiling point.

- (b) Explain your answer to (a). **O/C – P14 (2 marks)**

Water forms two hydrogen bonds per molecule.

Hydrogen bonding is the strongest intermolecular force. (1 mark)

Hydrogen sulfide, however, has dipole–dipole forces (and dispersion forces).

These are not as strong as hydrogen bonding. (1 mark)

Question 16 (3 marks)

- (a) What property of water enables an insect to walk on water? **O/C – P14 (1 mark)**

Surface tension

- (b) Explain the nature of the property of water identified in (a) in terms of intermolecular forces.
O/C – P7 (2 marks)

Water molecules at the surface form hydrogen bonding only with water molecules that are adjacent and below. There are no molecules (or attractions) from above. (1 mark)

These unbalanced forces on surface molecules result in a downward or inward force acting on surface molecules, ∴ surface tension. (1 mark)

Question 17 O/C – P4, 7 (3 marks)

You have already done an experiment where you used water's ability to absorb heat to measure energy changes in reactions.

- (a) Identify or describe the apparatus you used to perform the experiment. **(1 mark)**

Calorimeter or plastic foam cup + thermometer

- (b) Identify one measurement required to do this experiment. **(1 mark)**

- **Mass/volume of water/solution in the calorimeter.**
 - **Change in temperature of the water/solution in the calorimeter.**
 - **Mass of reagent involved in the reaction.**
- **Any one of the above.**

- (c) Write the equation you used to determine the amount of heat absorbed. **(1 mark)**

- **$q = m C \Delta T$**
 - **$\Delta H = - m C \Delta T$**
 - **$\Delta H = m C \Delta T$**
- **Any one of the above.**

Question 18 O/C – P4, 7, 10, 16 (5 marks)

A cargo helicopter accidentally dropped 1200 kg of chemical in a pond containing 50,000 litres of water. When the chemical dissolved in the pond, the temperature increased.

- (a) How much heat was released to the water in the pond, if the water temperature in the pond increased from 15°C to 21°C? **(2 marks)**

$$q = m C \Delta T = 50,000 \text{ kg} \times 4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1} \times (21^\circ\text{C} - 15^\circ\text{C}) = 1.254 \times 10^9 \text{ J} = \underline{1.3 \times 10^6 \text{ kJ}}$$

Consistent use of units = 1 mark

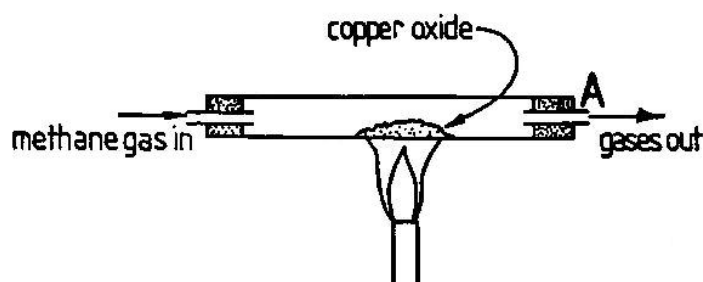
Correct calculation = 1 mark

- (b) Outline three implications (other than directly killing organisms) for aquatic life subjected to thermal pollution. **(3 marks)**

- Less dissolved oxygen causing stress to organism
- Increased metabolic rates which increases the demand for oxygen and so aggravates the low dissolved oxygen problem
- Fish eggs do not develop or hatch if temperature is too high
- False temperature cues are given to aquatic life,
∴ setting of migration and spawning at the wrong times of the year.

Question 19 O/C – P10, 13, 14 (5 marks)

The diagram shows methane gas passing over heated copper(II) oxide reacting to produce copper metal and gaseous products of carbon dioxide & water vapour which leave the apparatus at A...



- (a) Write the balanced formulae equation for the reaction of copper(II) oxide with methane (CH₄). **(1 mark)**



- (b) If 15.9 g of copper(II) oxide is reacted completely, calculate the mass of copper metal formed. **(2 marks)**

$$\text{moles CuO} = n = m \div M = 15.9 \text{ g} \div 79.55 \text{ g mol}^{-1} = \underline{0.200 \text{ mole}} \quad (1 \text{ mark})$$

$$\text{moles CuO} = \text{moles Cu}$$

$$\text{mass Cu} = m = n \times M = 0.200 \text{ mol} \times 63.55 \text{ g mol}^{-1} = \underline{12.7 \text{ g}} \quad (1 \text{ mark})$$

- (c) Calculate the percentage of copper in the copper(II) oxide sample. **(1 mark)**

$$\text{Cu}\% = \text{mass Cu} \div \text{mass CuO} \times 100 = 12.7 \text{ g} \div 15.9 \text{ g} \times 100 = \underline{79.9\%}$$

- (d) Calculate the volume of CO₂ produced at 25 °C and 100 kPa from 15.9 g of copper(II) oxide. **(1 mark)**

$$\text{moles CO}_2 = \frac{1}{2} \text{ moles CuO} = \frac{1}{2} \times 0.200 \text{ mol} = \underline{0.100 \text{ mol}}$$

$$\text{Volume CO}_2 = \text{moles} \times \text{molar volume} = 0.100 \text{ mol} \times 24.79 \text{ L mol}^{-1} = \underline{2.48 \text{ L}}$$

Question 20 O/C – P13 (4 marks)

- (a) Legislation states that the concentration of alcohol, $\text{C}_2\text{H}_5\text{OH}$, in the blood of an experienced driver of a motor car is not to exceed 0.05% (w/v).

Calculate the corresponding $\text{C}_2\text{H}_5\text{OH}$ concentration in the blood in moles per litre. **(2 marks)**

$$0.05\% \text{ (w/v)} = 0.05 \text{ g/100 mL} \quad (1 \text{ mark})$$

$$\text{moles ethanol} = 0.05 \text{ g} \div 46.068 \text{ g/mol} = 1.085 \times 10^{-3} \text{ mol}$$

$$\therefore [\text{C}_2\text{H}_5\text{OH}] = 1.085 \times 10^{-3} \text{ mol} \div 0.100 \text{ L} = \underline{0.01 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$

- (b) Identify a different measurement of concentration to those mentioned above (i.e. w/v) and describe a use for this measurement. **(2 marks)**

ppm (or ppb, v/v, w/w) (1 mark)

ppm is a convenient unit for measuring concentrations of pollutants, e.g. lead ions in waterways. (1 mark)

Question 21 O/C – P10, 14 (4 marks)

- (a) Calculate the mass of sodium sulfate required to prepare 50.0 mL of 0.150 mol L^{-1} solution. **(1 mark)**

$$\text{mol of Na}_2\text{SO}_4 = 0.15 \text{ mol L}^{-1} \times 50 \times 10^{-3} \text{ L} = \underline{0.0075 \text{ mol}}$$

$$\text{mass of Na}_2\text{SO}_4 = 0.0075 \text{ mol} \times 142.05 \text{ g/mol} = 1.065 = \underline{1.07 \text{ g}}$$

- (b) What volume of this solution would you need to dilute, to prepare 125 mL of $0.0500 \text{ mol L}^{-1}$ solution? **(1 mark)**

$$c_1V_1 = c_2V_2; V_1 = c_2V_2 \div c_1 = 0.0500 \text{ mol L}^{-1} \times 0.125 \text{ L} \div 0.150 \text{ mol L}^{-1} = 0.0417 \text{ L} = \underline{41.7 \text{ mL}}$$

- (c) What is the concentration of the sodium ions and sulfate ions in the $0.0500 \text{ mol L}^{-1}$ solution? **(2 marks)**



$$\therefore [\text{Na}^+] = 0.0500 \text{ mol L}^{-1} \times 2 = \underline{0.100 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$

$$\therefore [\text{SO}_4^{2-}] = \underline{0.0500 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$

Question 22 O/C – P1, 2, 13 (4 marks)

The table shows data for the compound hydrazine...

Composition	Hydrazine is a compound of nitrogen and hydrogen
Complete combustion of gaseous hydrazine at 400 K and 100 kPa	$\begin{array}{ccccccc} \text{hydrazine (g)} & + & \text{oxygen (g)} & \rightarrow & \text{nitrogen dioxide (g)} & + & \text{water (g)} \\ 1.0 \text{ L} & & 3.0 \text{ L} & & 2.0 \text{ L} & & 2.0 \text{ L} \end{array}$

- (a) Explain how the data for combustion illustrates Gay-Lussac's Law of Combining Gas Volumes. **(1 mark)**

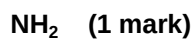
The gas volumes are in simple whole number ratios.

- (b) Determine the molecular formula of hydrazine.
Show all working. **(2 marks)**



$$\therefore x = 2; y = 4$$
$$\therefore \text{N}_2\text{H}_4 \quad \text{(1 mark)}$$

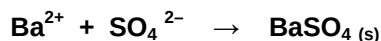
- (c) What is the empirical formula of hydrazine? **(1 mark)**



Question 23 O/C – P10, 13, 14 (7 marks)

2.08 g of barium chloride was dissolved in water to make 50.0 mL of solution and then added to 50.0 mL of an aqueous solution containing 2.84 g of sodium sulfate. A white precipitate formed.

- (a) Write the net ionic equation for the reaction forming the precipitate. **(1 mark)**



- (b) What is the mass of the precipitate formed? Show working. **(4 marks)**

$$\text{moles Ba}^{2+} = \text{moles BaCl}_2 = 2.08 \text{ g} \div 208.2 \text{ g/mol} = \underline{0.00999 \text{ or } 0.0100 \text{ mol}} \quad (1 \text{ mark})$$

$$\text{moles SO}_4^{2-} = \text{moles of Na}_2\text{SO}_4 = 2.84 \text{ g} \div 142.05 \text{ g/mol} = \underline{0.0200 \text{ mol}} \quad (1 \text{ mark})$$

$$\therefore \text{moles BaSO}_4 \text{ ppt.} = \underline{0.0100 \text{ mol}} \quad (1 \text{ mark})$$

$$\text{mass BaSO}_4 = 0.0100 \text{ mol} \times 233.37 \text{ g/mol} = \underline{2.33 \text{ g}} \quad (1 \text{ mark})$$

- (c) Calculate the concentration (mol L^{-1}) of sulfate ions in the final solution. Show working. **(2 marks)**

$$\text{moles SO}_4^{2-} \text{ excess remaining in solution} = 0.0100 \text{ mol} \quad (1 \text{ mark})$$

$$\therefore [\text{SO}_4^{2-}] = 0.0100 \text{ mol} \div 0.100 \text{ L} = \underline{0.100 \text{ mol L}^{-1}} \quad (1 \text{ mark})$$